

ARISTO EDUCATIONAL PRESS
HONG KONG DIPLOMA OF SECONDARY
EDUCATION EXAMINATION

(Mock Test 39)

Suggested Solution

Paper 1

1. D	11. C	21. D	31. C	41. D
2. B	12. D	22. D	32. D	42. D
3. A	13. B	23. B	33. C	43. B
4. D	14. C	24. A	34. A	44. D
5. A	15. A	25. C	35. B	45. C
6. C	16. C	26. D	36. C	
7. B	17. C	27. A	37. A	
8. C	18. A	28. B	38. A	
9. C	19. A	29. A	39. B	
10. B	20. B	30. B	40. D	

Explanation

1. Answer: D

If Irene visits Tokyo during a typhoon, she will be more likely to be injured, which may increase medical expenses.

2. Answer: B

(4) is incorrect. Interest is the compensation received by the lender for deferring consumption.

3. Answer: A

In a market economy, most resources are privately owned. Allocation of resources and distribution of goods are guided by market prices, i.e. by the market mechanism or price mechanism.

4. Answer: D

The firm is a private limited company.

Option D is correct. A private limited company is a legal entity and can sue other firms in its name.

5. Answer: A

A private good is rival in consumption. The amount available to others must be reduced once a person consumes the good.

Option B is incorrect. The price of a private good may be equal to zero, such as free gifts provided by companies.

Option C is incorrect. Apples are private goods because apples are both rival and excludable in consumption, not because the cost of producing an additional unit of apples is greater than zero.

Option D is incorrect. Private goods can also be provided by the government or individuals.

6. Answer: C

Since the percentage of workers engaged in tertiary production dropped from 79% to 75%, it means that the relative importance of tertiary production in terms of employment decreased.

Option A is incorrect. The question does not provide information on the number of workers in 2020 and 2024, so we do not know the changes in the number of workers engaged in primary production and secondary production.

Option B is incorrect. The question does not provide information on the employed population and the labour force in 2020 and 2024, so we do not know whether the overall employment rate remained unchanged.

Option D is incorrect. The question does not provide information on gross domestic product.

7. Answer: B

8. Answer: C

The provision of tuition subsidies for university graduates enrolled in full-time master's programmes will cause more graduates to choose to enroll in full-time master's programmes instead of looking for full-time jobs, so the labour supply will decrease.

Option A is incorrect. When the number of public holidays decreases, Country A's labour supply will increase.

Option B is incorrect. When the government has enacted legislation requiring companies to pay overtime to employees who work extra hours, the labour supply will increase.

Option D is incorrect. When other countries tighten immigration applications for Country A's residents, the number of people migrating to other countries will decrease, which will not reduce Country A's labour supply.

9. Answer: C

When the government builds gold storage facilities, it will help other related industries develop in Hong Kong. Hong Kong gold traders will have easier access to related services and the average cost of production will decrease. Gold traders in Hong Kong will enjoy external economies of scale.

10. Answer: B

The original profit-maximising output is 5 units.

Profit = Total revenue – Total variable cost – Total fixed cost

Option B is correct. If the market price decreases by \$2, the profit-maximising output will drop from 5 units to 3 units. The original total revenue is \$50 ($= \10×5), the new total revenue is \$24 ($= \8×3), and the total revenue will decrease by \$26. Total variable cost will decrease by \$19 ($= \$9 + \10). The profit will decrease by \$7.

Option A is incorrect. The question does not provide information on the marginal cost of the first 2 units of output, so the total cost of 5 units of output cannot be calculated.

Option C is incorrect. If the marginal cost for each unit of the output increases by \$2, the profit-maximising output will drop from 5 units to 3 units. The original total revenue is \$50 ($= \10×5), and the new total revenue is \$30 ($= \10×3), so the total revenue will decrease by \$20. Suppose the original total variable cost is \$x ($= MC_1 + MC_2 + \$8 + \$9 + \10). The new total variable cost is $x - 13$ [$= (MC_1 + \$2) + (MC_2 + \$2) + (\$8 + \$2)$]. Total variable cost will decrease by \$13. Profit will decrease by \$7 ($= \$20 - \13).

Option D is incorrect. Producer surplus = Total revenue – Total variable cost

An increase in total fixed cost will not affect the producer surplus of the firm.

11. Answer: C

When the transportation cost per kilogram of goods increases by 10%, the transportation cost per kilogram for all types of white rice will increase by the same amount. The relative price of high-quality white rice (in terms of average-quality white rice) will decrease. According to the law of demand, relative to average-quality white rice, people will buy more high-quality white rice. The average quality of imported white rice people purchase will increase.

Option A is incorrect. When the government abolishes the airport departure tax charged to each passenger, the relative price of long-distance flights (in terms of short-distance flights) will increase.

According to the law of demand, relative to short-distance flights, people will take fewer long-distance flights. The average flight distance will decrease. This is consistent with the law of demand.

Option B is incorrect. When the government imposes an *ad valorem* sales tax on each bottle of wine, the money price of all types of wine will increase and the quantity of wine sold (including high-quality wine) will decrease. This is consistent with the law of demand.

Option D is incorrect. This may be caused by an increase in the demand for air conditioners, which does not violate the law of demand.

12. Answer: D

13. Answer: B

Cross-border train services and air services are substitutes. An increase in the fare for cross-border trains will increase the demand for air services. The price and quantity transacted of air services will increase and total expenditure will increase.

Option A is incorrect. When the government reduces the basic allowance for the salaries tax, people's tax payment will increase and their post-tax income will decrease. Since air services are normal goods, demand for air services will decrease. The price and quantity transacted will decrease and total expenditure will decrease.

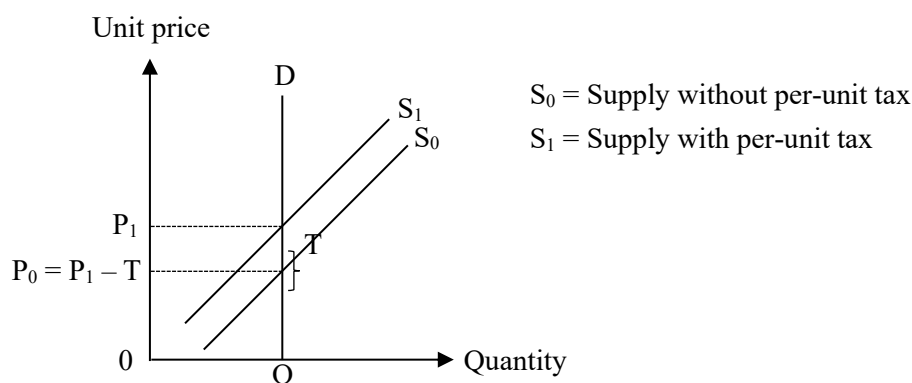
Option C is incorrect. As more airlines enter the market, the supply of air services will increase. The price will decrease and quantity transacted will increase. Total expenditure may increase, decrease or remain unchanged, depending on the price elasticity of demand.

Option D is incorrect. When the price of fuel increases, the cost of providing air services will increase. The supply of air services will decrease. The price will increase and quantity transacted will decrease. Total expenditure may increase, decrease or remain unchanged, depending on the price elasticity of demand.

14. Answer: C

Producers' total revenue net of tax = Price actually received by producers $\times$ Quantity transacted after tax
 $= (\text{Price after tax} - \text{Per-unit tax}) \times \text{Quantity transacted after tax}$

Option C is correct. If the demand is perfectly inelastic, after a per-unit tax is imposed, the equilibrium price will rise by the amount of the per-unit tax, the price actually received by producers will remain unchanged and the quantity transacted will also remain unchanged. The total revenue net of tax will remain unchanged.



Option A is incorrect. If the supply is perfectly inelastic, after a per-unit tax is imposed, the equilibrium price will remain unchanged. The price actually received by producers will decrease, while the quantity transacted will remain unchanged. Total revenue net of tax will decrease.

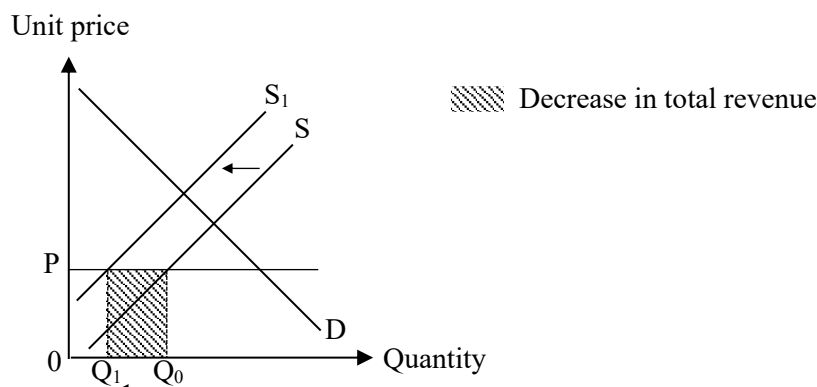
Option B is incorrect. If the supply is perfectly elastic, after a per-unit tax is imposed, the equilibrium price will rise by the amount of the tax. The price actually received by producers will remain unchanged, while the quantity transacted will decrease. Total revenue net of tax will decrease.

Option D is incorrect. If the demand is perfectly elastic, after a per-unit tax is imposed, the equilibrium price will remain unchanged. The price actually received by producers will decrease, and the quantity transacted will decrease. Total revenue net of tax will decrease.

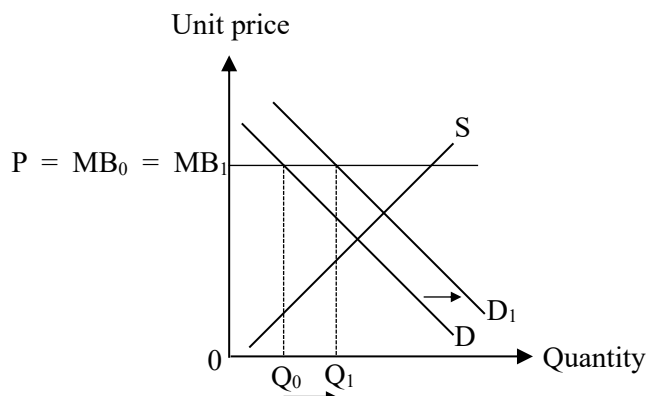
15. Answer: A

(1) is correct. If the price ceiling is effective, it means that the price ceiling is lower than the equilibrium price. An increase in the price of its complements will cause the demand for good X to decrease and the equilibrium price of good X will decrease. If the new equilibrium price is lower than the price ceiling, the price ceiling will become ineffective.

(2) is correct. If the price ceiling is effective, the quantity transacted will equal the quantity supplied. When the price of a factor of production increases, the supply of good X will decrease. The quantity transacted will decrease, while the price will remain unchanged. The total revenue of producers will decrease.



(3) is incorrect. An increase in people's preference for good X will cause demand to increase. If the price floor is effective, the marginal benefit of the last unit of good X will remain unchanged.



16. Answer: C

(2) is correct. In a perfectly competitive market, all buyers and sellers make consumption and production decisions based on a given price. The firm can sell as much as it wants at the prevailing market price and it has no influence on the market price. Therefore, the revenue from selling an additional unit of output, i.e. marginal revenue, is equal to the price.

(4) is correct. In a perfectly competitive market, since all sellers sell homogeneous products at the given market price and all buyers and sellers have perfect information, it is totally unnecessary for sellers to engage in non-price competition.

(1) is incorrect. In a perfectly competitive market, although individual firms cannot influence the market price, the market price will still change due to changes in market demand and/or market supply.

(3) is incorrect. In a perfectly competitive market, price is the only competitive criterion in determining the distribution of goods. There is price competition in a perfectly competitive market.

17. Answer: C

Quantity (units)	Original marginal benefit (\$)	New marginal benefit (\$)
1	10	12
2	8	10
3	6	8
4	4	6
5	2	4

The original market price is \$6 and the consumer will buy 3 units. The consumer surplus is equal to \$6 [= (\$10 + \$8 + \$6) – \$6 × 3].

The new market price is \$4. Under the new marginal benefit schedule, the consumer will buy 5 units.

The consumer surplus is equal to \$20 [= (\$12 + \$10 + \$8 + \$6 + \$4) – \$4 × 5].

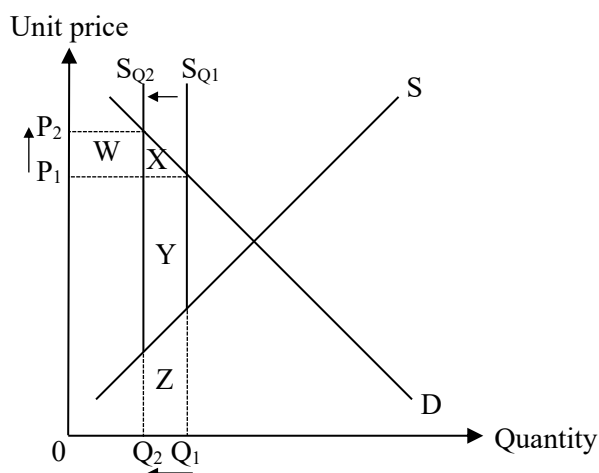
Consumer surplus will increase by \$14.

18. Answer: A

(1) is correct. After the government lowers the quota limit on good X, the market price will increase and the quantity transacted will decrease. The consumer surplus will decrease (by area (W + X)).

(2) is correct. The marginal cost curve of good X remains unchanged and the quantity transacted decreases. Total variable cost will decrease (by area Z).

(3) is incorrect. The gap between marginal benefit and marginal cost will widen, and deadweight loss will increase (by area (X + Y)).



19. Answer: A

(1) is correct. When the supply of good X is perfectly inelastic, after the per-unit subsidy is provided, the consumer surplus will not change, and the increase in producer surplus will be equal to the total amount of the subsidy. The per-unit subsidy will not cause deadweight loss.

(2) is correct. After providing the per-unit subsidy, the market price and quantity transacted of good X will remain unchanged. Consumers' total expenditure on good X will remain unchanged.

(3) is incorrect. The market for good X is in equilibrium and there is no excess supply.

20. Answer: B

(1) is correct. If the government imposes fines on the factory that discharges sewage, the marginal private cost of the factory discharging sewage will increase and the divergence between private cost and social cost will decrease.

(3) is correct. If the factory provides compensation to the affected fishermen, the external cost of pollution can be internalised. The amount of sewage discharged will be reduced to a level closer to the efficient level. Deadweight loss will decrease.

(2) is incorrect. If the deadweight loss due to excessive discharge of sewage is smaller than the maximum total social surplus, prohibiting the factory from discharging any sewage into the river will reduce the total social surplus from some positive number (the maximum total social surplus minus the deadweight loss due to excessive discharge of sewage) to zero. The total social surplus will decrease.

21. Answer: D

(1) and (2) are correct. They may explain why the average salaries of junior staff and supervisory staff in the financial services sector are higher than those in the construction sector.

(3) is correct. The higher academic qualifications of supervisory staff mean that they possess more human capital, which may explain why the average salary of supervisory staff in both industries is higher than that of junior staff.

22. Answer: D

Generally, the original income of persons with disabilities is lower. Increasing the amount of the personal disability allowance can reduce the tax they pay, and their post-tax income will increase, reducing the Gini coefficient based on post-tax post-social transfer monthly household income in Hong Kong.

Option A is incorrect. Tobacco tax is a regressive tax, which may increase the Gini coefficient based on post-tax post-social transfer monthly household income in Hong Kong.

Option B is incorrect. This will reduce the number of people eligible to receive CSSA and may increase the Gini coefficient based on post-tax post-social transfer monthly household income in Hong Kong.

Option C is incorrect. This will reduce the number of elderly people eligible to receive Old Age Living Allowance, which may increase the Gini coefficient based on post-tax post-social transfer monthly household income in Hong Kong.

23. Answer: B

Compared to the rich, the poor do not get more resources to improve their productivity or increase their future income, so distributing the same amount of cash to all permanent residents will not help equalise opportunities.

Option A is incorrect. The same amount of money will increase the income of low-income people by a larger percentage, which can reduce income inequality.

Option C is incorrect. When people's income increases, demand for inferior goods will decrease.

Option D is incorrect. The amount distributed is a transfer payment and will not be included in the calculation of Macau's government consumption expenditure.

24. Answer: A

$\text{GDP at market prices} + \text{Net factor income from abroad} = \text{GNI}$

The part of GNI at market prices that exceeds GDP at market prices represents the value of net factor income from abroad. The difference between GDP at market prices and GNI at market prices has narrowed, indicating a decrease in the net factor income from abroad.

Option B is incorrect. In 2023 and 2024, Country A's GDP at market prices was smaller than GNI at market prices, indicating that the net factor income from abroad was positive.

Option C is incorrect.

$\text{GDP at market prices} = \text{GDP at factor cost} + \text{Indirect taxes} - \text{Subsidies}$

In 2023 and 2024, Country A's GDP at market prices was smaller than GDP at factor cost, indicating that subsidies were greater than indirect taxes.

Option D is incorrect. The part of GDP at factor cost that exceeds GDP at market prices is subsidies minus indirect taxes. It can be seen from the figures that the difference decreased. This could be caused by a decrease in subsidies, or an increase in indirect taxes, or both.

25. Answer: C

Suppose 2023 is the base year.

(1) is incorrect.

$\text{Nominal GDP in 2023} = \$9 \times 12 + \$6 \times 28 = \276

$\text{Nominal GDP in 2024} = \$8 \times 14 + \$5 \times 26 = \242

Nominal GDP in 2024 is lower than that in 2023.

(2) is correct.

$\text{Real GDP in 2023} = \$9 \times 12 + \$6 \times 28 = \276

$\text{Real GDP in 2024} = \$9 \times 14 + \$6 \times 26 = \282

Real GDP in 2024 is higher than that in 2023.

(3) is correct.

$\text{GDP deflator in 2023} = 100$

$\text{GDP deflator in 2024} = (\$242 / \$282) \times 100 = 85.82$

The GDP deflator in 2024 is lower than that in 2023, reflecting a decrease in the general price level.

26. Answer: D

People's spending on consumer goods affects people's current living standard. Consumer goods constituting a smaller proportion of Country A's GDP compared to that of Country B means that the per capita nominal GDP may overestimate the general living standard of people in Country A.

Option A is incorrect. Per capita nominal GDP already considers the effect of the size of population on the general living standard of people.

Option B is incorrect. The Gini coefficient of Country A is lower than that of Country B (i.e., the income distribution of Country A is more even than that of Country B). This does not explain why the general living standard of people in Country A is lower.

Option C is incorrect. If Country A has more unreported production activities than Country B, Country A's per capita nominal GDP should underestimate the general living standard of its people.

27. Answer: A

When the long run aggregate supply increases while the aggregate demand remains unchanged, the new long run equilibrium point may be at A.

When the technology for production improves, the productive capacity of Country A will increase. Long run aggregate supply will increase while aggregate demand will not change.

Option B is incorrect. More people choosing to retire early will lead to a decline in the labour force, Country A's productive capacity will decrease and long run aggregate supply will decrease.

Option C is incorrect. A decrease in the price of raw materials will reduce the cost of production of firms. Short run aggregate supply will increase while long run aggregate supply will not be affected. The long run equilibrium point will remain at E.

Option D is incorrect. When the government increases transfer payments, people's disposable income will increase and private consumption expenditure will increase. Aggregate demand will increase while long run aggregate supply will not be affected.

28. Answer: B

When a typhoon hits a country and disrupts the supply of raw materials, the short run aggregate supply will decrease. The general price level will increase and the purchasing power of money will decrease.

Option B is correct. When banks tend not to lend out excess reserves, the deposit creation ability of the banking system of the country will decrease. The money supply will decrease and the interest rate will increase. Consumption/investment will decrease and aggregate demand will decrease. In the short run, the general price level will decrease and the purchasing power of money will increase.

Option A is incorrect. When people's desire to save decreases, private consumption expenditure will increase and aggregate demand will increase. In the short run, the general price level will increase and the purchasing power of money will decrease further.

Option C is incorrect. When firms expect the general price level to increase, they will reduce current supply. Short run aggregate supply will decrease further.

Option D is incorrect. When more people go to work overseas, the country's labour force will decrease and the short run aggregate supply will decrease further.

29. Answer: A

Licensed banks can accept deposits of any amount and maturity.

Restricted licence banks can accept deposits of any maturity of HK\$500,000 or above.

Deposit-taking companies can accept deposits of HK\$100,000 or above with an original maturity of at least three months.

Only licensed banks can accept Cindy's deposit of HK\$80,000, so institution B is a licensed bank.

Since the maturity of John's deposit is less than three months, institution A is not a deposit-taking company. Institution A may be a licensed bank or a restricted licence bank.

30. Answer: B

Monetary base = \$10 billion, reserves = \$3 billion, cash in public circulation = \$(10 – 3) billion = \$7 billion

Money supply M2 = Cash in public circulation + All types of deposits with licensed banks + NCDs issued by licensed banks held outside the banking sector = \$(7 + 32 + 5) billion = \$44 billion

Option C is incorrect. Demand deposits (\$12 billion) are already included in all types of deposits with licensed banks (\$32 billion), and thus should not be counted twice.

31. Answer: C

Required reserve ratio = [(Actual reserves – Excess reserves) / Bank deposits] × 100%
= [\$(400 – 50) million / \$1,400 million] × 100% = 25%

After the central bank issues \$200 million of newly printed banknotes, the monetary base will increase by \$200 million. Since the public always hold \$20 million in cash, the cash in public circulation remains unchanged, representing an increase of \$200 million in bank reserves. New bank reserves is \$600 million.

Required reserves = Actual reserves – Excess reserves

The new excess reserves are \$100 million, the new required reserves = \$600 million – \$100 million = \$500 million

New bank deposits = \$500 million × (1/25%) = \$2,000 million

The new balance sheet is as follows:

Assets (\$ million)		Liabilities (\$ million)	
Reserves	600	Deposits	2,000
Loans	1,400		

Option C is correct.

Actual banking multiplier = $1 / \text{Actual reserve ratio}$

Original actual reserve ratio = $(\$400 \text{ million} / \$1,400 \text{ million}) \times 100\% = 28.57\%$

Original actual banking multiplier = 3.5

New actual reserve ratio = $(\$600 \text{ million} / \$2,000 \text{ million}) \times 100\% = 30\%$

New actual banking multiplier = 3.33

The actual banking multiplier will decrease.

Option A is incorrect.

Original money supply = Cash in circulation + Bank deposits = \$20 million + \$1,400 million = \$1,420 million

New money supply = \$20 million + \$2,000 million = \$2,020 million

The money supply increased by \$600 million.

Option B is incorrect.

Original loans = \$1,000 million

New loans = \$1,400 million

Loans increased by \$400 million.

Option D is incorrect.

Maximum banking multiplier = $1 / \text{Required reserve ratio}$

As the required reserve ratio remains unchanged, the maximum banking multiplier will not change.

32. Answer: D

Policy I: Tightening the requirements for consumers opening a stored value facility account will lead to an increase in transaction demand for money, so money demand will increase. Raising the discount rate will reduce the money supply. To conclude, the nominal interest rate will rise.

Policy II: Changing civil servants' salary payment from monthly to weekly will reduce the transaction demand for money, so money demand will decrease. The nominal interest rate will decrease. Raising the required reserve ratio will cause the money supply to fall and the nominal interest rate will increase. To conclude, the change in the nominal interest rate will be uncertain.

Policy III: After introducing the dividend tax, people will tend to reduce the proportion of shares in their asset portfolios, so asset demand for money will increase and money demand will increase. Selling government bonds in the open market will reduce the money supply. To conclude, the nominal interest rate will rise.

33. Answer: C

34. Answer: A

Quantity theory of money: $MV = PY$, where M = money supply, V = velocity of circulation of money, P = price level and Y = real output.

Assume V is constant.

From year 1 to year 2, the percentage increase in the money supply was smaller than the percentage increase in real output, so the percentage change in the price level was negative, i.e. the general price level decreased.

Option B is incorrect. According to the quantity theory of money, the velocity of circulation of money is assumed to be constant.

Option C is incorrect. As the percentage change in the price level was negative, the inflation rate was negative.

Option D is incorrect. According to the quantity theory of money, the percentage change in nominal output (PY) will be equal to the percentage change in the money supply. As the percentage increase in the money supply was smaller than the percentage increase in real output, the percentage increase in nominal output was smaller than the percentage increase in real output.

35. Answer: B

There was unexpected deflation in Country A in 2023.

The real value (purchasing power) of the fixed rental paid by the tenants would increase, so they would lose.

Option A is incorrect. The real value (purchasing power) of the interest received by holders of fixed rate bonds would increase. They would gain.

Option C is incorrect. The interest rate would be adjusted in accordance with the inflation rate, so the purchasing power of the repayment received by the banks would not be affected.

Option D is incorrect. When there was unexpected deflation, the purchasing power of cash would increase, so people holding cash would gain.

36. Answer: C

Under a progressive tax system, as high-income earners pay a higher average tax rate than low-income earners, income distribution will become more even. Under a regressive tax system, as high-income earners pay a lower average tax rate than low-income earners, income distribution will become more uneven.

Option A is incorrect. As taxable income increases, the tax payment of a regressive tax may increase, decrease, or remain unchanged.

Refer to the following examples:

Taxable income (\$)	Tax payment (\$) (figures in brackets are the average tax rates)		
	Tax system 1	Tax system 2	Tax system 3
1,000	100 (10%)	100 (10%)	100 (10%)
2,000	120 (6%)	90 (4.5%)	100 (5%)

Tax systems 1, 2 and 3 are all regressive, with the average tax rates falling as taxable income increases. Therefore, when taxable income increases, the tax payment under a regressive tax may increase.

Option B is incorrect. Only a proportional tax is in line with Adam Smith's equity principle of taxation.

37. Answer: A

When a deflationary (output) gap exists in an economy, an increase in aggregate demand or short run aggregate supply can help the economy restore to its long run equilibrium.

Option A is correct. This will encourage firms to increase investment. Investment expenditure will increase and aggregate demand will increase.

Option B is incorrect. Having the central bank purchase government bonds from the public is a monetary policy, not a fiscal policy.

Option C is incorrect. Raising the standard rate of the salaries tax will reduce people's disposable income. Private consumption expenditure will decrease and aggregate demand will decrease.

Option D is incorrect. Imposing a per-unit sales tax will reduce the short run aggregate supply.

38. Answer: A

The effect of an expansionary monetary policy on real output is affected by the deposit creation ability of the banking system. When the deposit creation ability of the banking system decreases, the effectiveness of an expansionary monetary policy on real output will decrease.

(1) is correct. If more loans lent out by commercial banks are not redeposited into the banking system, but are held in cash by the public, the deposit creation ability of the banking system will decrease and the effectiveness of an expansionary monetary policy on real output will be reduced.

(2) is correct. If companies are less willing to obtain loans from banks, demand for loans may be insufficient. The deposit creation ability of the banking system will decrease and the effectiveness of an expansionary monetary policy on real output will be reduced.

(3) is incorrect. If commercial banks are more willing and able to lend out their excess reserves, the deposit creation ability of the banking system will increase.

39. Answer: B

After the construction is completed, Hong Kong's productive capacity will increase and the long run aggregate supply will increase.

It will promote commercial, tourist and leisure activities. Private consumption expenditure, investment expenditure and net exports (due to an increase in exports of services) will increase, and aggregate demand will increase.

40. Answer: D

When the government reduces the unemployment benefits, government expenditure will decrease and the fiscal balance will improve. The disposable income of unemployed people will decrease, consumption (part of which is spent on imported goods and services) will decrease and the trade balance will improve.

Option A is incorrect. This will reduce imports of goods and improve the trade balance. However, this has no significant impact on the fiscal balance.

Option B is incorrect. When the central bank lowers the discount rate, the money supply will increase and the interest rate will decrease. Consumption/investment will increase, and aggregate demand will increase. In the short run, the output level (national income) will increase, government tax revenue will increase and the fiscal balance will improve. However, imports will also increase as the output level increases, and the trade balance will worsen.

Option C is incorrect. When the government increases the business registration fee, government revenue will increase and the fiscal balance will improve. However, this has no significant impact on the trade balance.

41. Answer: D

Depreciation refers to the reduction in the value of fixed capital as a consequence of its use in a period of time. It cannot be negative in value.

Option A is incorrect. When total output decreases as the quantity of a variable factor increases, marginal product will be negative.

Option B is incorrect. When total revenue is smaller than total cost, profit will be negative.

Option C is incorrect. When the actual inflation rate is greater than the nominal rate of return from holding bonds, the actual real rate of return from holding bonds will be negative.

42. Answer: D

Option A is incorrect. The question does not provide production information for Country A's trading partners, so we cannot determine whether Country A has an absolute advantage in producing good X.

Option B is incorrect. Comparing the total output with no international trade with the total output with specialisation and international trade, Country A produces an additional 20 ($= 30 - 10$) units of good X and gives up 40 units of good Y.

Country A's opportunity cost of producing 1 unit of good X $= 2 (= 40 / 20)$ units of good Y

Option C is incorrect. Comparing the total consumption with international trade with the total output with specialisation and international trade, Country A exports 10 units of good X in exchange for 25 units of good Y. The terms of trade is 1 unit of good X $= 2.5$ units of good Y or 1 unit of good Y $= 0.4$ units of good X.

Option D is correct. Country A's gain from trade of exporting 1 unit of good X $= 2.5$ units of good Y $- 2$ units of good Y $= 0.5$ units of good Y

Country A's gain from trade of exporting 10 units of good X $= 0.5$ units of good Y $\times 10 = 5$ units of good Y

43. Answer: B

When the demand for wheat increases, the domestic price of wheat will increase. Domestic production will increase, and the total revenue of domestic producers will increase.

Option A is incorrect. The domestic price will increase.

Option C is incorrect. A change in demand in a small open economy will not affect the international price. It will remain at P_w .

Option D is incorrect. The import volume will remain unchanged at the quota limit.

44. Answer: D

The economy recorded a trade surplus in 2021, meaning that the total value of exported goods was greater than the total value of imported goods. From 2021 to 2024, the year-on-year percentage change in the total value of exported goods was greater than the year-on-year percentage change in the total value of imported goods, indicating that the increase in the total value of exported goods was greater than the increase in the total value of imported goods, so the trade surplus was increasing.

Option A is incorrect. The question does not provide information about exports of services and imports of services, so we cannot determine the change in the invisible trade balance.

Option B is incorrect. From 2021 to 2024, the year-on-year percentage change in the total value of exported goods was positive, indicating that the total value of exported goods was increasing.

Option C is incorrect. The question does not provide information about services, external factor income flows and current transfers, so we cannot determine the change in the current account balance.

45. Answer: C

After deducting the handling fee, Mr. Cheung could exchange HKD 100,000 for approximately CAD 17,491 ($= 100,000 \times (1 - 1\%) / 5.66$) on 31 October 2023.

On 31 October 2024, Mr. Cheung exchanged the amount back to Hong Kong dollars. After deducting the handling fee, he could get approximately HKD 96,624 [$= 17,491 \times (1 - 1\%) \times 5.58$]. Mr. Cheung lost approximately HKD 3,375.

Paper 2

Section A (44 marks)

1. (a) The quantity of labour will change as output changes. (1 mark)
 (b)

Machines (units)	Labour (units)	Average product (units)	Total product (units)	Marginal product (units)
3	2	5	10	–
3	3	4	12	2
3	4	4	16	4
3	5	5	25	9
3	6	6	36	11

No. (1 mark)

The data does not show that marginal product eventually decreases. (1 mark)

2. (a) Verbal elaboration:

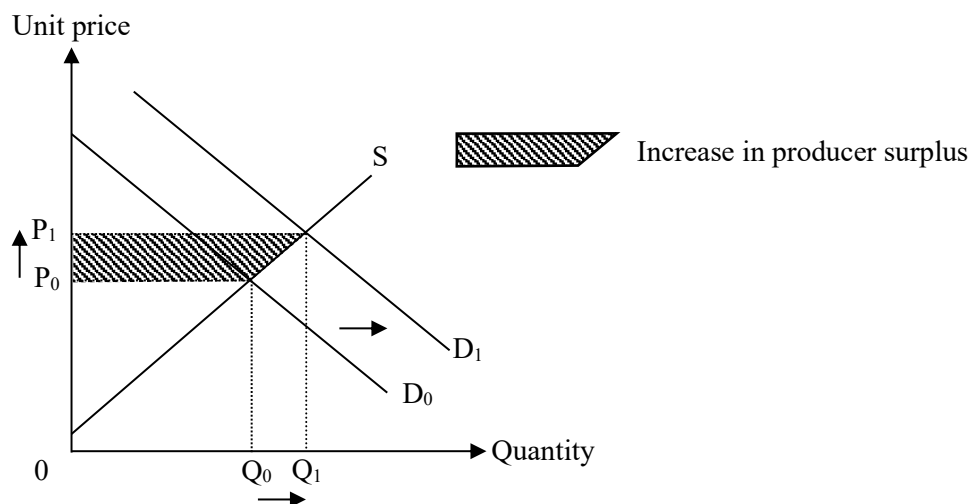
When the United States imposes an import tariff on motor vehicles produced in Canada, the supply of motor vehicles produced in Canada will decrease and the price will increase. Motor vehicles produced in the United States and those produced in Canada are substitutes, so the demand for motor vehicles produced in the United States will increase. (2 marks)

The producer surplus will increase. (1 mark)

Illustrate in the diagram:

- rightward shift of demand curve (1 mark)

- producer surplus increases (1 mark)

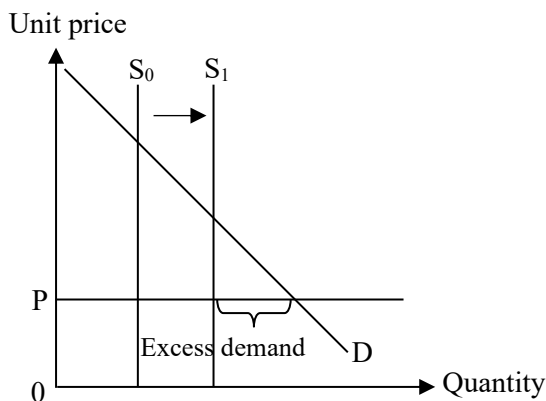


- (b) No. (1 mark)

The resale price of second-hand imported cars will increase and the driver's opportunity cost of continuing to use his car will increase. (2 marks)

3. (a) Illustrate in the diagram:

- vertical supply curve (1 mark)
- rightward shift of supply curve (1 mark)
- price is below the equilibrium level (1 mark)
- correct position of the excess demand/shortage (1 mark)



(b) Expected real interest rate = Expected nominal interest rate – Expected inflation rate = 4% – 2% = 2%

Actual real interest rate = Actual nominal interest rate – Actual inflation rate = 4% – 3% = 1%

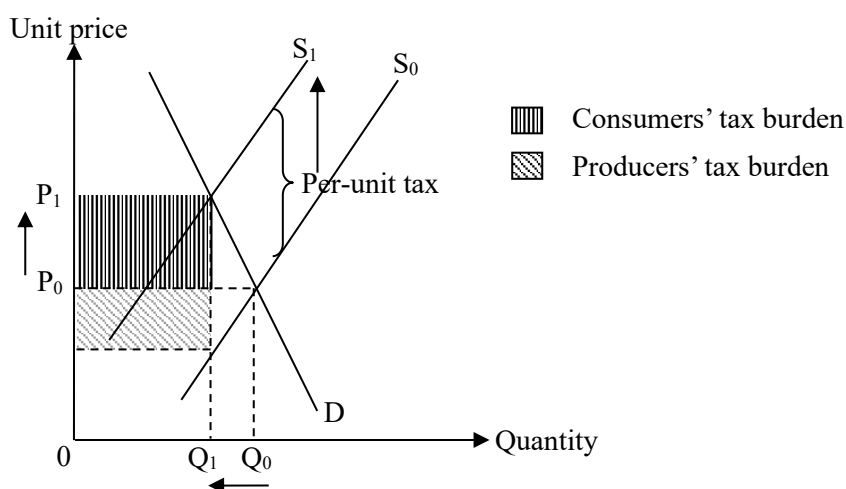
As the actual real interest rate is lower than the expected real interest rate, the real value of the interest he receives will be lower than expected and he will lose. (3 marks)

4. Verbal elaboration:

The condition is that the supply of high-sugar foods is more elastic than its demand. (2 marks)

Illustrate in the diagram:

- upward shift of supply curve (1 mark)
- price increases (1 mark)
- quantity transacted decreases (1 mark)
- correct positions of consumers' tax burden and producers' tax burden (1 mark)
- consumers' tax burden > producers' tax burden (1 mark)



5. Reasons:
- Not all the inputs used to produce garments originated from Hong Kong, for example, some raw materials were imported.
 - Some of the garment were not produced in the current year, for example, some of them came from inventories.
 - Any reasonable answers
- } 2@, max: 2 marks
- (Mark the **FIRST TWO** points only)
6. (a) Excess reserves = \$750 million – \$3,100 million × 20% = \$130 million (1 mark)
- Possible reasons for keeping excess reserves:
- Banks may deliberately keep more reserves than required to prepare for sudden withdrawals of funds by a large number of depositors.
 - There is insufficient demand for loans.
 - Any reasonable answers
- } 1@, max: 2 marks
- (Mark the **FIRST TWO** points only)
- (b) Cash in public circulation = \$(900 – 750) million = \$150 million (1 mark)
- Money supply = \$(150 + 3,100) million = \$3,250 million (1 mark)
- (c) New required reserve ratio = 20% + 5% = 25% (1 mark)
- Maximum possible amount of deposits after deposit contraction = \$750 million × (1/25%) = \$3,000 million (1 mark)
- Maximum possible change in deposits = \$(3,000 – \$3,100) million = –\$100 million (1 mark)
7. (a) Country A's opportunity cost of producing 1 unit of garments = 0.5 (= 1 / 2) units of rice (1 mark)
- Country B's opportunity cost of producing 1 unit of garments = 0.67 (= 2 / 3) units of rice (1 mark)
- Since Country A's opportunity cost of producing 1 unit of garments is lower than that of Country B, Country A will export garments. (1 mark)
- (b) Total output of garments in Country A = 30 (= 30/1) units of garments (1 mark)
- Total output of rice in Country B = 10 (= 30/3) units of rice (1 mark)
- (c) Range of terms of trade: 1.65 units of garments < 1 unit of rice < 2 units of garments (2 marks)

Section B (60 marks)

8. (a) Oligopoly. (1 mark)
- Features:
- several dominant sellers
 - firms are interdependent in pricing strategy
 - price rigidity
 - any reasonable answers
- (Mark the **FIRST TWO** points only)
- (b) An advantage:
- Part of the business risk can be shared by the pilots. If the demand for air services decreases, wages paid to the pilots will be lower.
 - Any reasonable answer
- (Mark the **FIRST** point only)
- (c) When firm D was closed, the number of substitutes for air services provided by the other airlines would decrease, so the price elasticity of demand for the air services they provide would decrease. (2 marks)
- (d) No, as the shareholders of a listed company/public limited company enjoy limited liability, Miss Cheung's liability is confined to the amount of her investment in firm D. (2 marks)
9. (a) In the short run, the adjustment of input prices (e.g. wages and rent) is imperfect or slow. As a result, when the general price level increases, output prices will increase faster than input prices, making it profitable for firms to produce more output. At the same time, the imperfect adjustment of input prices leads to a fall in the real cost of production, so firms will increase their inputs to produce more output. Therefore, the SRAS curve is upward sloping, i.e. a higher price level would result in a higher output level. (3 marks)
- (b) The economy has an inflationary (output) gap because the output level is higher than the full-employment output level. (2 marks)

(c) Verbal elaboration:

When the government raises the standard tax rate of the salaries tax, people's disposable income will decrease and private consumption expenditure will decrease. (2 marks)

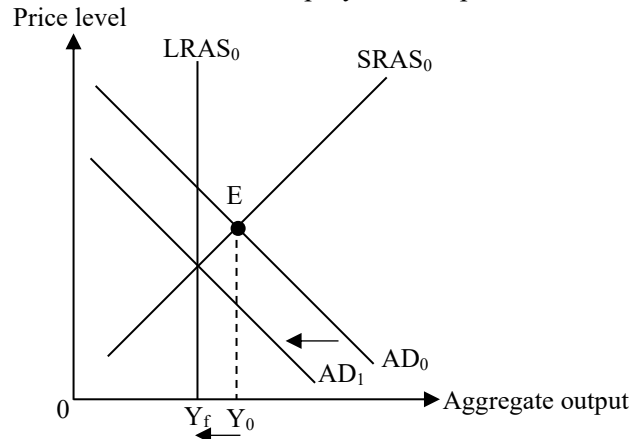
Aggregate demand will decrease. (1 mark)

Aggregate output will decrease, returning to the full-employment output level. (1 mark)

Illustrate in the diagram:

- leftward shift of AD curve (1 mark)

- aggregate output decreases to the full-employment output level (1 mark)



(d) The government can increase the sales tax rate to restore aggregate output to the full-employment output level. (1 mark)

Sales tax is a regressive tax. Increasing the sales tax rate will widen the income gap between the rich and the poor in the economy and the income distribution will become more uneven. (1 mark)

Only high-income earners need to pay the standard tax rate. Increasing the standard tax rate of the salaries tax will narrow the income gap between the rich and the poor in the economy and the income distribution will become more even. (1 mark)

(e) A monetary policy:

- an open market sale of government bonds
- an increase in the discount rate
- an increase in the required reserve ratio
- Any reasonable answer

1 @, max: 1 mark

(Mark the **FIRST** point only)

The monetary policy will cause the money supply to decrease and the nominal interest rate to increase. (1 mark)

10. (a) No, (1 mark)
 as the crops are neither sold to the market nor added to inventory, they are not included in the calculation of gross domestic product. (1 mark)
- (b) (i) No. (1 mark)
 Inflation is a persistent increase in the general price level. A once-and-for-all increase in the price of a specific type of product after the typhoon does not constitute inflation. (1 mark)
- (ii) A function of price:
 - Rationing function: The food is distributed to the highest-valued buyers.
 - Signal function: Prices help transmit information about the supply of food relative to the demand for food.
 - Any reasonable answer
 (Mark the **FIRST** point only)
- } 2@, max: 2 marks
- (c) Hong Kong's current transfers to abroad will increase and Hong Kong's balance of payments will be worsened. (2 marks)
- (d) No, (1 mark)
 as the time Ms. Chan spent on the volunteer work may have alternatives uses, an opportunity cost may be involved. (1 mark)
11. (a) Advantages:
 - obtain information about the general public from the government more easily for making business decisions
 - can provide stable services at a lower price
 - obtain loans more easily with government support
 - any reasonable answers
 (Mark the **FIRST TWO** points only)
- } 1@, max: 2 marks
- (b) The difference between the expenditure and the underlying income of the WKCD Authority has narrowed, and the amount of loss has decreased. (2 marks)
 OR
 The increase in the WKCD Authority's underlying income was greater than the increase in the expenditure, and the amount of loss has decreased. (2 marks)
- (c) The WKCD creates employment opportunities in other tourism-related supporting facilities, so the WKCD Authority creates a benefit to other parties, but they do not give any compensation to the WKCD Authority. External benefit thus exists and there is a positive externality. (3 marks)
- (d) Unemployment rate = (Unemployed population / Labour force) × 100%
 If the WKCD Authority reduces the number of positions for foreign professionals, and they all return to their places of origin, Hong Kong's labour force will decrease. As the unemployed population will not change, the unemployment rate will increase. (4 marks)

(e) Marks awarded for effective communication (EC: max 2 marks)

Marks	Performance
2	• Arguments supported with the source/data and appropriate economic theories • Relevant materials presented • Well-organised and coherent answers presented without repetition of ideas • Ideas clearly and fluently expressed with appropriate use of language/words/terms/symbols
1	• Arguments presented with some support of the source/data and economic theories • Some irrelevant materials presented • Answers presented in a less organised way with some repetition of ideas • A clear message conveyed with some inappropriate use of language/words/terms/symbols
0	• Arguments presented with no support of the source/data and economic theories • Materials unrelated to the gist of the question presented • Inconsistent arguments presented • Limited ideas expressed with inappropriate use of words/terms/symbols

The maximum marks for content are 12 marks. Answers may include the following:

How suggestion I may achieve its objective:

- With reference to the view of the accountant in Source D, the WKCD Authority will receive a huge one-off income from selling the residential developments to private property developers. With reference to the view of the current affairs commentator in the same source, the income of the WKCD Authority includes bank interest. The WKCD Authority can deposit the amount from the sale of residential developments into banks to generate more interest income.

How suggestion II may achieve its objective:

- Reducing the number of employees and lowering their wages can reduce the total variable costs of running the WKCD, and profit can be increased.

Possible factors that may limit the effectiveness of suggestion I:

- Private property developers' demand for residential developments is derived from people's demand for residential properties. With reference to the view of the accountant in Source D, if the residential property market is in a downturn, demand for residential properties will decrease, and private property developers' demand for residential developments will also decrease. The revenue generated from selling the residential developments will be lower and the interest income that can be generated will decrease accordingly.
- The interest income that can be generated depends on the deposit interest rate. If the deposit interest rate decreases, the interest income that can be generated will decrease.

Possible factors that may limit the effectiveness of suggestion II:

- If reducing the number of employees and lowering their wages lead to a decline in the overall service quality of the WKCD (such as the types of exhibits collected, environmental cleanliness, types of museum merchandise, etc.), the revenue of the WKCD Authority (such as ticketing income) may drop. If the reduction in revenue is greater than the reduction in total variable cost caused by reducing the number of employees and lowering their wages, profit will decrease.
- In terms of reducing expenditure, the effect of reducing the number of employees and lowering their wages on total variable cost may not be significant compared to the total cost of operating the facilities in the WKCD (such as the expenses of acquiring collections for museums and marketing expenses), so profit may not be increased effectively.

Effects of suggestion I on Hong Kong's macroeconomic variables:

- Private property developers building residential properties in the WKCD can increase Hong Kong's investment expenditure, and aggregate demand will increase. In the short run, the output level and the price level will increase.
- Private property developers may profit from residential developments, leading to an increase in the government's profits tax revenue.

Effects of the suggestion II on Hong Kong's macroeconomic variables:

- If the laid-off employees cannot find jobs or if the wages of the employees are lowered, the salaries tax revenue of the government will decrease.
- If the laid-off employees cannot find jobs, the government may need to provide transfer payments to them (such as Comprehensive Social Security Assistance), and government expenditure will increase.

Section C (16 marks)

12. (a) Illustrate in the diagram:

- correct position of Q_M

(1 mark)

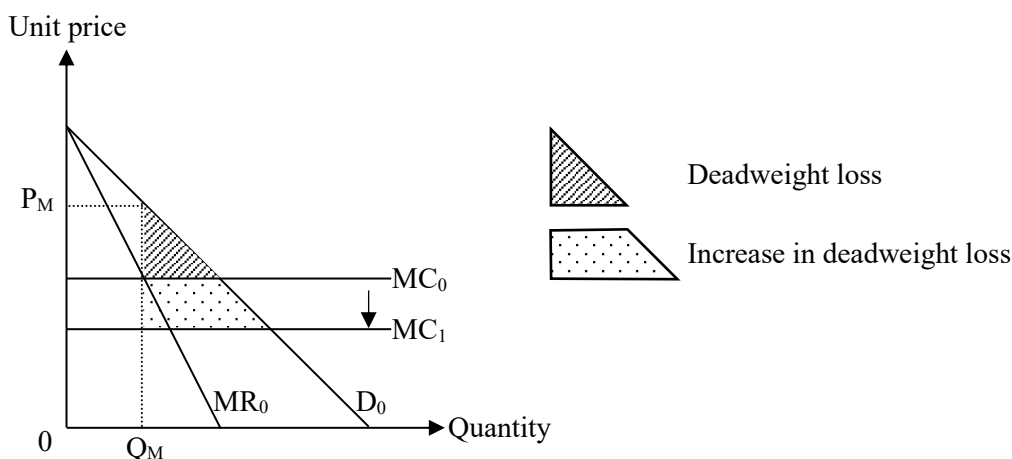
- correct position of P_M

(1 mark)

(b) Illustrate in the diagram:

- correct position of the deadweight loss

(1 mark)



(c) Verbal elaboration:

Economic efficiency would be worsened and the deadweight loss would increase because the gap between marginal benefit and marginal cost would be widened and the output would be further away from the efficient level.

(2 marks)

Illustrate in the diagram:

- downward shift of the MC curve

(1 mark)

- correct position of any of the following: increase in deadweight loss/new deadweight loss/increase in the gap between marginal benefit and marginal cost

(1 mark)

(d) (i) Yes. This is because customers with different data usage will end up paying a different average price for the same service produced at the same cost.

(3 marks)

OR

No. This is because all customers are entitled to the same pricing arrangement and there is no market segmentation.

(3 marks)

(ii) Yes. Different groups of customers (new customers and existing customers) are charged different prices for the same services produced at the same cost.

(3 marks)

(e) (i) Tie-in sales.

(1 mark)

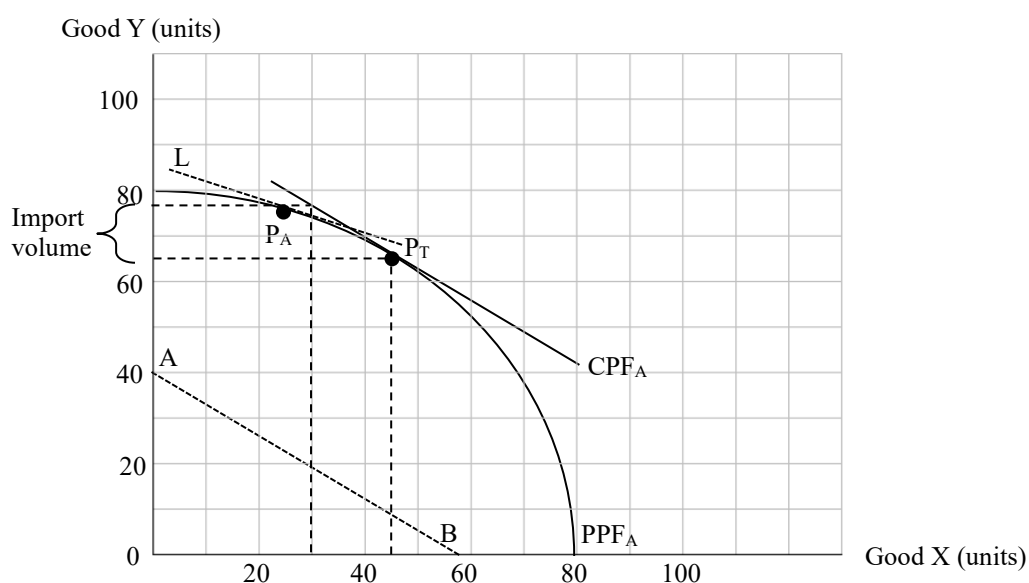
(ii) The second conduct rule,

(1 mark)

because an abuse of market power may be involved.

(1 mark)

13. (a) In Country A, the marginal costs of producing both good X and good Y are increasing. (1 mark)
- (b) Verbal elaboration:
- The slope of the straight line (L) is the marginal cost of producing good X. (1 mark)
 - L is flatter than AB. / The marginal cost of producing good X is lower than the international price of good X. (1 mark)
 - Country A has a comparative advantage in producing good X. (1 mark)
- Illustrate in the diagram:
- correct position of the straight line (L) (1 mark)
- (c) (i) Illustrate in the diagram:
- correct position of the production point after trade (P_T) (1 mark)
 - tangency between CPF_A and PPF_A (1 mark)
 - CPF_A is parallel to the dotted line AB (1 mark)
- (ii) Illustrate in the diagram:
- correct position of import volume (1 mark)



(d) (i) Possible reasons:

- People may leave school earlier to participate in international trade, reducing the mean years of schooling.
- After engaging in international trade, Country A's pollution may become serious, shortening people's life expectancy at birth.
- Any reasonable answers

2@, max: 4 marks

(Mark the **FIRST TWO** points only)

(ii) A developmental dimension:

- income distribution
- environmental sustainability
- Any reasonable answer

1@, max: 1 mark

(Mark the **FIRST** point only)

(e) A policy:

- provide subsidies for imported medical goods to increase people's life expectancy at birth
- provide (more) free education to extend people's mean years of schooling and expected years of schooling
- Any reasonable answer

2@, max: 2 marks

(Mark the **FIRST** point only)